



General Sir John Kotelawala Defence University

Department of Computer Science

CS11021– Programming Laboratory (Intake 43)

Lab Sheet 11

1. Develop a C++ function that simulates a **mobile banking app login**, displaying a welcome message after a user enters their username.
2. Develop a C++ function that generates an **online shopping invoice**, taking the product name, quantity, and unit price as arguments, and displaying the total cost.
3. Develop a C++ function that simulates **activating a smart fan** when room temperature exceeds a certain threshold.
4. Develop a C++ function that displays a **flight departure announcement**, including the flight number and departure gate.
5. A user wants to calculate the **total electricity bill** based on units consumed and cost per unit. Implement this using a function. (Return Type)
6. A gym member wants to calculate their **monthly membership fee** after applying a discount percentage. Implement this using a function. (Return Type)
7. A company needs to calculate an employee's **annual income with allowances and deductions**.
 - Return Basic Salary from the function to the main program.
 - Calculate Final Annual Income in main().
8. A temperature monitoring system needs to convert **Fahrenheit to Celsius**.
 - Return Fahrenheit temperature from the function to the main program.
 - Convert to Celsius in main().

9. A student loan system needs to calculate the **total payable amount and monthly repayment**.
- Return Loan Amount from the function to the main program.
 - Calculate total amount with interest & monthly installment in main().
10. A vehicle tracking system needs to compute **average speed, estimated travel time, and fuel cost**.
- Return Distance from the function to the main program.
 - Calculate Speed, Time, and Fuel Cost in main().
11. A wildlife research team is tracking the movement of a wild animal in a forest reserve.
- Write a function to return the **distance traveled by the animal in a day** (based on input values).
- In main(), calculate:
- Total distance traveled over multiple days
 - Average daily movement
 - Display whether the animal is **highly active or less active** based on a threshold
12. Due to a fuel shortage, a driver must carefully plan a journey.
- Write a function to return the **available fuel in liters**.
 - In main(), calculate:
 - Maximum distance that can be traveled (based on vehicle fuel efficiency)
 - Check if the planned trip distance is possible
 - If not possible, display how much **additional fuel is required**